ActInf GuestStream #010.1 ~ Philip Gerrans

GuestStream | Philip Gerrans | 2:00:43

hello everyone welcome to the active
inference lab today it's september 9th
2021
for guest stream number 10.1 double
digits or double digits with a decimal
and we're really honored and
appreciative to have professor philip
guerins today
coming in from australia and also dave
joining us on stream
we're gonna have a presentation from
philip and then we're gonna have some
time for discussion and dialogue so if
you're watching live please feel free to
leave questions in the live chat and we
will address those on the stream
and we're just really looking forward to
this and again appreciative that you're
coming on our stream so thanks and take
it away phillip
okay great uh thank you very much thank
you very much for the invitation i spoke
to anna kanika who told me that she had
a great experience with you guys so i'm
very very happy to be able to
present and to tell you a little bit
about the background to this talk i was
a
originally a philosopher of mind and of
cognitive science and i'm very
interested in
uh the
use of
models and techniques in cognitive
science to understand psychiatric

disorder
and i found predictive processing very
helpful when i was writing a book on
this a few years ago but looking back at
um some other psychiatric disorders like
the cotard delusion and so on
i think it's really great to kind of go
a little bit deeper and more more
fundamental if you think of predictive
processing as kind of like a
computational structure at the level of
representation and algorithm
but of course a lot of psychiatric
disorders i think you've got a very sort
of a bottom-up genesis
we really need to know about how the
brain is working to control the body and
even how the body is working in its
social world
so the active inference framework for me
has been very helpful
in rethinking this in this book that i'm
writing right now
and also i think it's it's actually
crucial to understanding a phenomenon
that i'm talking about right now that
the kotar delusion
the cotard delusion is so interesting
because um
well it's interesting in itself as
you'll see it's fascinating phenomenon
but it points us to some really
interesting uh
interesting facts or yeah interesting
evidence about the way the mind is mind
is working at a really basic level and
how that ramifies
and cascades upwards throughout

cognition and behavior so let's go i'll
i'll just tell you what
the kotar delusion is it's a delusion of
sometimes people think of it in the folk
world as
the delusion that you're dead
i was discovered in 1882
but
we described it more accurately i think
as a delusion of inexistence
so here's a here's someone who has it
she had the experience of having no
identity or self
and being only a body without content
she convinced that her brain had
disappeared and the whole body was
translucent another one more recent one
i speak and breathe and eat but i'm dead
okay so there's some classic
examples but here
is a really recent one published in the
last couple of years this is fascinating
cases coming from mexico
there's a really good group there that
works on this
so mr a was referred to the national
institute of neurology because of
psychotic and catatonic symptoms
his appetite diminished he barely slept
he stayed withdrawn and silent
he complained about his body saying his
tendons had dried up and his organs were
getting bigger speak of an internal
hemorrhage his organs have been torn
apart he began to insist that he was
dead i can't feel the blood flowing
through my veins
didn't deprive him of looking for sharp

objects to cut his neck and forearms
then he gradually improved first he
became mute began to urinate on his
clothes refused to eat
and
then speak
finally you know after some treatment he
said he's improved slightly but he said
i don't have feelings because i'm dead
he had a right-sided heart and that it
stopped
and then he kept on saying even after
persisting treatment his nihilistic
delusions as they're accurately called
you know delusions of inexistence of
nothingness i am dead among other human
beings it's like a program where i'm
dead
and he said he was a dead person among
the corpses under the earthquake debris
now
the genesis of this case of qatar
delusion is um
encephalitis a form of encephalitis
where it's an autoimmune reaction
against proteins in the brain so it's
one of the most basic
things that can go wrong at the cellular
level and if the protein is in methyl
the aspartate the condition is called
nmdar antibody encephalitis okay so
that's what mr a had but of course our
interesting question is
if we know what's happening you know at
the molecular level at the level of kind
of membranes and receptors
why on earth would that make you think
that you didn't exist

so they were addressing that they were
in the in the field of addressing the
question you know how does the brain
enable experience
and cognition that's that's the
questions of cognitive neuroscience
and psychology and anyone else who wants
to
get involved in that topic phenomenology
philosophy
okay but it's a really uh pertinent
question
so here's the outline of the talk i just
told you about the delusion and then i'm
going to tell you about some of the
classic attempts to explain it
uh just run through those
and point to a problem with those
and then in discussing uh one of the
most recent and persuasive and really
interesting
um explanations
uh that we owe to alexandre bien i'm
going to introduce the active inference
and i'm going to show how the active
inference approach
can
rectify a problem with beyonce's account
but also a point to point to some other
deep sort of unifying themes
across cognitive science and across
disorders so the active inference
framework i think gives a elegant
and persuasive
um way to explain not just you know
cases like mr a
but a lot of other dis other disorders
and pathologies where our experience of

our self
goes astray so things like pain and
symbolia where people say that they're
experiencing pain but also that their
body is experiencing pain but it's not
happening to them
but it feels as though it's not their
pain or depersonalization disorders in
which people will say not just that pain
is not theirs but
they feel outside of or detached from
their whole experience okay so and the
solution that i'll propose i'll give you
um just a short summary right now
is that as part of the um
project or process of of active
inference the body the brain makes uh
hierarchical models of itself and the
world of deep the deep structure of the
world
as as populated with objects and
properties that the external world with
which it interacts
but of course the brain also has to make
a model of its own
internal uh
uh
functioning
the processes that that keep it keep
keep us alive and functioning and in
order to do that it makes a model of
itself as a unified persisting entity
and that's what we feel the self that's
what we feel
when we feel
like like persisting people when we feel
like ourselves we we are the model
that's what we're experiencing activity

in the circuitry that makes that uh
self-model okay
and in a cotard delusion there's the
self-model
uh goes wrong in a certain way or that
process of self-modeling goes wrong in a
certain way but we can see other
disorders in which that process of
self-modelling goes wrong
and i think that
theory makes
it maps really nicely to the um the
neural substrates of all these disorders
it explains the symptoms and so on so
let's go
so of course as we saw um
such an interesting disorder the the
quartile delusion because
you know there's there's questions in
sort of philosophy and about
rational belief fixation and so on i
mean how could someone coherently say
i've died or i don't exist and there's a
question about
rationality there and the nature of
reasoning and so on on the basis of
experience which is interesting in
itself
which is that's part of the philosophy
of delusions i guess or the philosophy
of psychiatric disorder where i came
into this topic a while ago
but of course it's also very interesting
in okay how can you also have the
experience of not existing you know so
it points us to the question of what is
going on when you actually feel like a
self

okay and of course it's also a great
test case for theories that
try and explain experience and cognition
in terms of underlying
neural functioning so here's the kind of
broad brush of the kind of classic
approaches to the these topics that you
get in theories of delusion since i
guess
the early 90s
which tried to wed
philosophy of belief rational belief
fixation to the neuroscience of dilution
formation and that
the most top-level abstract way of
putting it is
there's two stages in delusion formation
the first one you know is an unusual
binomial as they say
sensory
experience so something goes wrong you
know at the level of sensory motor
functioning
gives you an unusual experience
and then the second stage is
a metacognitive response to that
experience so this is why
someone for example who
lost control or was unable to feel
uh
unable to feel that they are the agent
of their actions when they're picking up
things and moving around the world okay
that's the loss of it as they call sense
of agency
due to um you know normally problems in
the
parietal lobule inferior parietal lobule

something goes wrong there with the kind
of sensory motor
in action of agency and then the people
what person will say to explain that
weird experience of moving around the
world but not feeling like they are the
person who initiated the movement or
controlling it and say someone else is
controlling me you know so delusions of
control as they say that would be a
classic case we have an anomalous
experience
at the first stage and then the delusion
is a belief or some kind of
metacognitive rationalization or story
that the person tells okay
now
there's two implications of the um the
two-stage theory
one is that well if the patient doesn't
have
a problem with higher order processing
then she won't develop a delusion
because the delusion is a metacognitive
response right so in that case the
person should have
an anomalous experience she should feel
like
she's making movements that aren't hers
for example or you might feel like
you're hearing voice you might hear like
feel like you're hearing voices
and the person will just report that i
mean having an unusual weird
distressing experience but they wouldn't
go on to have a delusion because the
delusion is the result of a
problem at the stage of as they call

belief fixation or metacognition
okay
and if this is really right then there
should be cases in which people have the
unusual experience
but they don't go on to
have a delusion and of course we are all
familiar with this in everyday life if
you think about it
you've probably had uh feelings of deja
vu or jamaica you know in deja vu you
think i've been here before i've had
this experience before even though you
know that you haven't
okay and this is the point that you know
you can have an experience
that is completely anomalous or weird
don't go on to have a delusion about it
okay and in this case you say well it
feels as if i've been here before or in
the opposite case jamie vu
you know you walk into your bedroom and
it feels where you sleep every night but
you just have this fleeting instant of
uh
what this is new or this is feels like a
duplicate of my bedroom or something but
you don't go on to form a delusion
because rational insight is retained
unlike the classic case of delusions of
misidentification which i spend no time
on
in the delusion of misidentification
a person will say something like
they'll see their wife or see their
husband or boyfriend or something and
say they're a stranger see and you say
why do you say that you know don't they

do they look the same as their
appearance chant and then they look
identical
but they are like a replica or a
duplicate or an imposter in the classic
delusion
and the story is what has gone wrong
here is there's a problem at the level
of sensory processing
this thought is that normally when we
see a familiar face we get an affective
response to that face which is part of a
signal that that's a familiar person you
know i see you
i see my sister or my my partner and
then it's like oh yeah that's that's her
so it's a feeling of familiarity that's
based in affect but when that goes
missing
then all the other perceptual
information is intact i recognize the
phase the shape of the face and so on
but i don't they don't feel familiar and
in the case of
delusions of misidentification the
person rationalizes these losses or
alterations to the sense of familiarity
by saying oh
you know my wife's been replaced
by an imposter okay amazing delusions
but there you go
but that's just an example of the kind
of two-stage
theory of delusion formation which was
developed for delusions of
misidentification now in turning to the
quarter division
what is

the first stage within this framework
well alexander bion
says that
it's the loss of what people call the
sense of mindness
or uh subjective presence is what anil
seth calls this and people from sussex
who work on de-personalization
basic idea is that
something that philosophers and many
people have remarked on or
thought about
is that when you have experiences
part of that experience is an awareness
that you are the person who's having
experience as jasper's said
every psychic manifestation perception
bodily memory it carries with it this
feeling of being mine and having an eye
quality of personally belonging it's
been termed personalizations
and so the thought is just as when you
see familiar people there's this kind of
subtle um just this little subtle
feeling of familiarity which you don't
even notice that it's there until it
goes missing
and so it's the case that with your
everyday experience it feels like it's
yours
i mean who else so there's this kind of
aspect of subjectivity or subjective
presence
that you wouldn't even become aware of
until it went missing we only see it as
we only even notice that this is an
experience
a kind of subtle accompaniment to all of

life
until
we are confronted with as beyonce's
exceptional and pathological cases and
then if it goes missing you're really in
trouble
right so for example
in case of pain a symbolia
what can happen is that
someone can
have a pain they can have the experience
of pain
you know
um and they can report that experience
of pain saying yeah you know i mean yeah
i'm in pain um but it doesn't feel like
it's my pain
or i feel detached like it's not
happening to me
until you know when on interrogation you
say
yeah yeah you know it feels hot i feel
kind of pressure it feels intense in
other words
they can
be
still able to detect
and represent uh bodily damage through
the process of as it's called you know
nociception the primary representation
of pain stimuli
okay but something else has gone missing
right and what uh in this nice um
and there's two theories about penis
symbolia
one is that what's gone missing is an
emotional response to pain so that would
be the thought that

in the normal case of pain
you have no conception
you know the detection and
representation of bodily damage and then
there's if you like an affective or an
emotional response or interpretation to
that
and the resultant pain experience is
like a merger or combination bodily
damage plus
uh distress and that's what you report
but when you take the distress part out
you just get the representation of
bodily damage
that's one theory of pain
but colin klein had a different account
and on colin klein's view it's not a
loss of affect this is not a loss of
affect it's a loss of the sense of
mindness
so a symbolia on his view
is a localized
version of depersonalization experience
asymbolia is like a restricted case
just just to the pain processing
where the connection between uh the
processing of bodily damage or
nociception as it's called
and
subtle feeling that the experience
belongs to you comes apart so as he says
this sort of unity may fail
okay
and that this leads to what he calls
depersonalization or pain
as opposed to
depersonalization in the general case
where

the the loss of the sense of mindness is
not just for pain
but for everything
so someone with severe personalization
disorder
will
move through life
you know having sensations
sensations of movement you know
perceptual sensations bodily experiences
but feeling
as if those things don't belong to them
as someone says here it feels almost
like i have died but no one has thought
to tell me so i'm living in a shell that
i don't recognize
i felt my brain was somewhere else and
just watching me i couldn't tell whether
i was present or whether i was the part
that it was gone there was the one
watching and the other
it's like seeing life as if it were
played in a film or a movie but in that
case who is watching the film
i find myself regarding existence so
from beyond the tomb i'm outside my own
body and individuality i'm
de-personalized
so just to recap
okay klein gives us the idea that you
know symbolia you lose this this sense
of mindness or belonging to me for one
particular uh channel of bodily
processing that's um nociception or pain
signaling
but in depersonalization this loss is
global and then and when the loss is
global the person reports

depersonalization for everything and the
feeling there is like
well they say it feels as if i don't
exist it feels like i'm not there right
and so
how should we explain what's going on
here well there's different theories
uh beyond has one i had one 20 years ago
it's a bit out of date now i think i'm
updating it today
even as we speak
and then we have the one i'll be talking
about uh today which uh we owe to the
active infant skies like seth sakirus
and so on
but here's beyond's take on it which is
very interestingly
so
dion uh what beyonce says
is that depersonalization is this case
of an anomalous experience and the
anomalous or strange experience is the
feeling that the self
is unreal
and it's not there
and so if we go back to the cotard
delusion remember the two-stage theory
of delusion is that
a delusion arises when you have an
anomalous experience
and then you have some
problem at a higher level of
rational interpretation narrative
organization or metacognition
so that you uh
you have no rational insight into the
experience or are unable to
explain it or you end up just taking it

for granted
and that's beyond's idea about
depersonalization it's the first stage
of cottage delusion
so when someone with depersonalization
says it feels as if i don't exist it
feels as if i'm not here it feels as if
experience is outside me or i'm not
involved in it
that's because they're having the same
experience as the kotard person had the
kotard patient has but they have
rational insight and intact
metacognition so they can look they can
as it will look at that experience and
say
it feels as if i don't exist
it feels as if experience isn't mine
but
i know that's not true so i report it
just saying it feels that way but i
don't they don't develop the delusion so
what differentiates cotard and
depersonalized delusions
is that the kotard patient has the same
experience
as a um as a patient with
depersonalization
but they have intact rational insight
okay
now
um that's the philosophical angle but
recently
this
very interesting
group of guys in mexico
have a neuropsychiatric version
of um alexandra beyond's two-stage

account which is they explain mr a's
case you know the man who thought that
his organs had dried up and that he was
dead
he has insular cortex dysfunction and
that insular cortex dysfunction is the
first factor
okay so the insular cortex dysfunction
produces
the experience
of depersonalization the feeling that
that one sensory experiences and
cognitions don't belong
to the subject
or that
and then
there's a prefrontal cortex dysfunction
that produces the second phase the
second factor so this is the story
if you look on the left you've got uh
the um the first stage
insular cortex
um produces this anomalous subjecting
feeling state so if you have hypoactive
insular cortex or maybe lesion or you
know neurochemical disruption
then you're going to have insular cortex
problems
and that's going to lead to the feeling
that experiences are not yours in some
way that could manifest as pain a
symbolia depersonalization
or it's also in alexa thymia which we
didn't talk about today
but in any case that gives rise to
this abnormal sensory state or if they
call it abnormal perception so something
has gone wrong with your ability to feel

that experiences belong to you
thanks to your insular cortex
dysfunction
and then
if you also have
prefrontal cortical dysfunction then you
can't explain that you have no rational
insight into that and so you end up with
the delusion
now just to recap the um the assumption
here that's shared by beyond
and by ramirez bermudez and his
collaborators
is that the first stage is essentially
the same or similar between the
personalization experiences and qatar
delusion
okay
so the anomalous experience of
depersonalization with rational insight
well that's depersonalization
but if you have the anomalous experience
of depersonalization with no rational
insight then you get the cotard delusion
right
and the result and this is because of
hypoactivity or lesion maybe in your
insular cortex
but this is this is where i want to go
today this is wrong right this doesn't
fit with the phenomenology of cotard
syndrome or the neurobasis of cotard
syndrome which is typically
a profound and catastrophic
reconfiguration of the bodily and
effective life world of the patient it's
not a restricted quasi-perceptual
deficit

and it's not just um insular cortical
dysfunction
um because look look at the variety of
conditions in which you see cottage
delusion
just some right affective disorders
schizophrenia syphilis cephalo
encephalitis hemorrhage
malformations brain neoplasm the cancer
migraine parkinson's disease dementia
right this wide spread
you know
catastrophic
uh
conditions of you know neural
malfunction and then and then you see um
a quater delusion it's not just the ins
not just the insular cortex
um hypoactive okay
and not only that it's not just
cartilage delusion doesn't look like a
i'm just like oh it feels as if i'm not
here
if you go back to the initial
description of the symptom of
symptomology it's not just i feel like i
have
intact perception and cognition it just
feels like it's it's not mine or i feel
detached from it i mean typically some
of qatar delusion undergoes like
completely profound
and uh quite catastrophic
reconfiguration of their whole life
experience they feel like their body is
expanding or constructing or you know
that guy felt as though his organs had
dried up

they can't sleep they stay urination and
sexual function and everything goes to
um crazy their appetite is out of luck
you know it's a complete
catastrophic breakdown
uh often it's progressive but there's no
doubt about it that's a catastrophic
breakdown
okay
so there's a first one we can disagree
with the foundational assumption but at
the same token but at the same time
sorry at the same point
it is true that insular malfunction is
um associated with cottage with with uh
depersonalization i beg your pardon
so let's look at that first
and it's also true that um the insulin
malfunction among other things is
profoundly involved in cataract dilution
and i think we can see why and this is
where active inference
makes its appearance enter stage right
active inference theory
uh active inference and predictive
processing provide this really nice
framework to um tie all these ideas
together to explain depersonalization
cotard asymbolia
and to fit together the kind of neural
correlates the phenomenology and
symptomatology symptomology okay
well first let's let's start with the
kind of
the restricted thing let's talk about uh
depersonalization why would insular
damage
lead to depersonalization why would it

be the case that if you had hypoactivity
in your insular cortex
you would feel like your experience
wasn't happening to you
it's pretty interesting and amazing
uh sensation well the reason is that the
insular cortex
it does a lot of things you know as they
say it's an enigmatic
um and ubiquitously involved neural
structure it's very kind of deep it's
kind of it receives afferents from
almost everywhere it communicates with
almost every every part of the brain
okay and you'll see it involved in in so
many things
so it's hard to it's always been very
hard to work out a particular function
that the insular cortex does right it's
not like you know for example
looking at part of the visual cortex
that you know processes
wavelength information or something like
that or is involved in you know
these neural circuits are involved in
determining movement or trajectory or
something like that it's not like that
the insular cortex somehow seems to
contribute to so many things as a as an
integrative um
device of integration and i would say as
a device of organization and
coordination and it's like a relay
station actually between all these other
different things
but the posterior insular cortex is very
important in somatosensory vestibular
and motor integration

okay
so that the posterior insular cortex is
basically involved i would say in
integrating all the information about
what's happening
in your body right
while the anterior insular cortex which
is our real culprit here has pres
projections to limbic regions
which are you know that's one of the
substrates of emotional processes
and is associated with integrating
autonomic and visceral information in
other words the basic information about
how your body is functioning
um you know my blood pressure my blood
acidification oxygenation you know
state of my organs and so forth this
kind of really basic keeping you alive
uh kind of information integrating that
with emotional cognitive and
motivational functions
in other words and here's the killer it
brings internal signals into
consciousness and emotional
interpretations supporting the neural
constructions of
subjective feelings
so um just to go
let me stay here for a moment
so the thought you know that we get with
people like bud craig and everything
like that is that the anterior insular
cortex
basically takes all these signals about
what's happening on the inside on the
inside of your body you know my blood
pressure's up my heart is beating

whatever my organs are functioning well
or not
and integrates that with all the other
uh cognitive systems that are telling
you what's going on outside you know
what you see what you hear what you
smell and the higher cognition what you
remember what you mentioned what you're
thinking about
okay and that creates it integrates all
this in other words to kind of to allow
you to to
create if you like a representation of
how what is happening in the world
is interacting with what's happening
inside
my body to enable me to function
adaptively in context
and as they say well what is that except
a sense of self a sense of sense of self
or representation of their self is
i'm the thing that has to manipulate my
body around the world to interact and
engage adaptively with it and to do that
i need to integrate all the information
that's relevant to that task that's what
we need a sense itself for and that's
what the anterior insula does
anterior insulator looks down at the
bodily processing looks out to the world
and integrates them to give you a sense
of as they say subjective feeling in the
moment
now so with that in place we can see why
when um
anterior insula goes wrong you or you
know malfunctions you might get a uh
the sense of not being involved in

experience
now the most um
persuasive i think explanation of how
this might go wrong is that it's a
dissociative response
to trauma this comes from medford's here
and david the people who work on it
but they explain it in kind of emotional
terms so they call it they call a result
as emotional numbing so the thought is
you're having a traumatic experience
you're being raped or being bashed or
maybe you're in a
you know destructive abusive environment
and
as a way of coping you know your
your
your brain shuts down the anterior
insular cortex it just turned it just
turns it off and what is the result you
still end up and you still end up you
know seeing feeling thinking
and so on but you don't have the kind of
effective as i say effective response or
as as uh sierra puts it
you no longer have the emotional
coloring of experience
so it leads to a qualitative change of
conscious awareness which is reported by
the subject as unreal or detached so the
basic idea is you shut down
affective processing
so that you can cope with
extreme distress or trauma
and to shut that down you shut off the
anterior insular cortex so you no longer
feel
that it's happening to you so you can

see what you have here is really an
effective or an emotional theory of
self-awareness the basic idea is
self-awareness is really created by
the emotional coloring of experience
and when that is gone you end up in a
emotionally monochrome world so to speak
and there's some evidence that lack of
anterior insulin insulin is related to a
diminished emotional responsiveness seen
in dpd
and that when if the insulin comes back
online um not only they restore their
effective responses you know
um
then they start to feel that they are
this feeling of detachment and
disengagement disappears
i just make the point that um this
theory of um
this theory this inhibitory theory of
depersonalization is dissociative it
actually
it really requires a predictive
processing framework because it's not
just the fact for example that
the subject doesn't feel effective
responses it is the fact that
it doesn't feel effective responses in
contexts or conditions where they are
predicted where they should occur you
know so depersonalization arises when
you know i don't know take take a mother
who's just given birth
and then someone gives her the child to
cuddle or nurse
and she doesn't feel anything
right or you come home from a long trip

your children run up and jump into your
arms and give you a big hug you don't
feel anything
or you know you hear bad news um you
hear terrible news someone close to you
has died or whatever
don't feel anything now it's not just
that you don't feel anything that's the
problem it's that your mind predicts
that you should your mind
has this kind of ongoing model
of the world and you and you in the
world that your progress through the
world and your interactions with the
world will be as they say emotionally
colored
and these these emotional colorings are
effective responses which get you around
the world
and when that's gone
right it's not just that it's gone so
you know because
we have plenty of experiences
that are emotionally neutral walking
down the street you know i don't know
doing your tax return um reading a book
you're not especially interested in
all that kind of jazz listening to a
talk you know going going to a
department meeting or that is a slightly
distressing experience but you know you
know what i mean the mere fact that
there's no effective response doesn't
make you feel depersonalized it's the
loss of affective response in a
condition where your mind predicts it
it's that it's the um
it's the disjunction between or the the

contradiction between
there should be affect and now there
isn't
and that's reported by then that's the
thought is i feel like i'm not here okay
and so the hypothesized mechanism is
that the
you in a traumatic situation
and it's too much for your brain to deal
with so
these inhibitory processes kick in
to shut down the affective response in
the moment as a dissociative response
and its
inhibition of the anterior insular
cortex by this inhibitory structures
okay
and this is exactly what i just said
why that produces depersonalization is
because affective experience is
important in um
[Music]
in getting around life and also it's
predicted you know your mind's model of
you acting in the world is that you will
have it
uh
have uh
effective responses to the world
here's where the affective inference um
starts to kick in okay so then the
question is
how does the mind predict
that your progress through the world
will have
or will give rise to effective
experiences how does it do that
well i think it does so in a process of

what is called
uh inactive inference framework
predictive processing framework
hierarchical self-modeling
okay
and basically hierarchical self-modeling
is as i said before briefly that um
just as the mind you know makes uh
models or representations of the
external world the world out there
in terms of its causal structure
then the mind has to make a model of
what's going on inside the body in terms
of its causal structure
and the way the mind does this is by
integrating all the properties and
features of the world and attributing
them to unified persisting causes well
that's objects in in the case of the
external world that unified persisting
causes are objects you know you don't
see
smears of colors and movements and
changes you see cars driving down the
railway you don't see these kind of
weird interacting shapes and properties
you see people's faces smiling and so on
so
it's kind of obligatory strategy for the
mind to integrate
properties and features by attributing
them to
underlying
objects or in the case of substances if
you want to call it that unified
persisting things
all right and that's what that's what we
do in outward and the idea of the higher

of the self-modeling framework is that's
what we do inward as well the mind's
obligatory strategy is to unify and
integrate when there are regular
coherent
predictable
flows and sequences of
experiences or sensory information
the mind explains that if you like
interprets it and predicts it by
attributing it to the same object
this happens at
multiple levels right at the most basic
level you know you have to integrate all
these different different homeostatic uh
functioning you know blood pressure
uh visceral dilation you know vascular
dilation all this kind of stuff that
keeps you going
well actually you don't
that is actually integrated
into by attributing all of those things
as occurring in the same entity and we
feel the result
in what in an interceptive state
interception being the internal analogue
of perception so example which makes
this point well i think is
uh fatigue right well
when you are fatigued
right what do you feel you feel like i'm
fatigued the the thing you know the
embodied thing feels depleted
but you don't feel
uh all these individual streams of you
know glycogen depletion adenosine
depletion um slow metabolism all that
you just get the kind of overall vibe of

i'm tired
right and that's good because then you
can go to sleep you don't have to fine
tune
anyway you can't you can't get in there
and change your glycogen levels or
anything like that you can just eat so
hunger is another example of a kind of
interceptive sensation that is um
you know this does this kind of
aggregative and integrative thing
and
this kind of type of self modeling this
is very familiar from the predictive
process from from that
takes place at multiple levels given the
kind of imperatives
that we have to survive
okay so let me just spend a little bit
of this okay at the most within the um
active inference
framework
okay the most basic task
of any organism self-organizing system
is to survive it has to maintain its um
maintain its physical integrity you know
its boundary with the world
um it has to take in energy from the
world you know it has and so on has to
engage with the world take in energy
from the world and then it has to do
something with that energy to maintain
itself
to keep all its components interacting
so that's at the most basic physical
level organisms avoid
this inevitable tendency to
thermodynamic decay that we all face and

you know the probabilistic and analog of
that is they avoid entropy they have to
keep themselves
in a state far far from entropy by
taking in energy and using it
of course at the biological level what
does that actually mean that means
staying alive surviving prospering in
the world
and the most when we look at the
physiology
okay
then obviously we're talking about
homeostatic mechanisms we're talking
about mechanisms of self-maintenance
for a um
for a
discrete you know unique individual so
to speak now there can be obviously
cases where um
borderline cases of home is homeostasis
where you know
where organisms are dependent on other
organisms for example there are lots of
cases where there's a kind of you know
criss-crossing of boundaries but let's
keep it simple for the moment
think of a cell or think of an organism
got to maintain their their um
basic functioning and that's what
referred to as homeostasis or allostasis
slightly larger
more sophisticated concept
so these are the kind of basic physical
imperatives but of course to do that
there has to be a lot of um signaling
within the system you know the levels of
levels of endocrine function levels of

blood pressure levels of blood
oxygenation they are signaled
to a brain or a protobrain or a mind if
you want to call it that
and that signaling okay
that signaling requires
that the organism
in some sense implement or maintain a
model which predicts
what is um
what is uh optimal for it how far it's
departing from optimal and whether the
actions that it's taking to get back
into
optimal range
are
successful
and when it can do that
when it can successfully model itself
then it's minimizing what they call
variational free energy
and surprise here just refers to
departure from
the predicted optimal state given the
organism's action now if that sounds too
abstract just think you know here's your
brain
um and you're riding up you're starting
to ride your bike up a hill or walk
upstairs okay and you have a kind of
good
optimal range of heartbeat function and
blood oxygenation okay but that needs to
change now because you're walking
upstairs right you need to you need more
oxygen in your blood you need to work
harder and raise your heartbeat
so your mind you know sends if you like

brain sense instructions
raise heartbeat you know this is all at
the kind of
signaling level it's quite dynamic
if your heartbeat goes up that's good it
should be up when you're going upstairs
and it should be up if you're riding
your bike up a hill so you've minimized
you know free energy you've stayed
within your optimal bound okay and
you're not getting in any information
that's um out of
out of the kind of predicted um
yeah that violates a prediction my model
for that situation
but
let's say that you come back downstairs
and you sit down and you're resting
and having a drink
and your heartbeat is still racing well
that's outside optimal range that's not
what your um
your body and mind you know predict for
that situation your brain
think your brain doesn't well your brain
doesn't think but yeah now there are
signals saying this is out of kilter i'm
sitting on my couch my heartbeat is 200
or rather i'm sweating sweating
furiously but i'm not riding my bike
okay and it's not 40 degrees outside
what's going on i've got a so a lot of
signaling going on all the time
about whether and how you are optimizing
your bodily function and so that's just
that the level of signaling and some of
that's very very dynamic and leads to
these kind of automatic adjustments but

of course at a higher level when you
talk about um
cognition or what you can think of the
mind or as what we call cognition as the
construction of sort of representational
or computational systems you know
that exploit these information or
properties or that organize these
informational and signaling properties
so that we end up with a
representational architecture that's how
we get perception
memory sensory motor control and so on
and what
that's what the theory of computation
gives us for understanding the mind it
gives us
what the theory of computation basically
is
in its most abstract is turning
information
information and signaling into
representation representations that can
be manipulated okay translate these
abstract ideas
into the idea of
self-modeling
and we get something
like this we get okay at the very
basic level
of organismic function
then you have allostatic or homeostatic
integration your mind is kind of your
brain is looking around
am i in or out of am i in or out of
optimal functioning all this signaling
is going on and what we call
interception

is the integration of those processes
okay which attributes them to the thing
which is being regulated the thing which
is being managed okay the thing in which
all of these um
processes have to cohere to enable
adaptive functioning and then we get a
readout of that and the readout is your
interceptive feeling fatigue you know if
you feel tired
that's because you know glycogen
depletion adenosine levels have changed
you know metabolism is slow blah blah
blah muscles are weaker and and not
contracting as fast
and you feel that as fatigue as its
integration of all of those
allostatic variables and that happens
basically the job of the posterior
insular cortex written there as pi
is to do that that's where all that
stuff coheres right that's where all of
that stuff coheres
then of course there's not just the
issue of um
how is my body functioning
that's one thing
okay is my body de-energized well yeah
that's important to know
but then
there's a separate kind of we might call
a high level issue of the meaning and
significance of those bodily processes
for you
in context right so if your heartbeat is
racing
and you're riding your bike up a hill or
going for it in the gym

that's a good thing right that shouldn't
cause you any particular distress you'll
feel that as exhaustion
but if your heartbeat is racing because
you're waiting for your cancer test
you know and the doctor has just come
back just come out of his office and
he's looking at the um
he's looking at your chart and he's
looking at your um he's looking at your
ultrasound with a look of uh concern and
worry
now your heart is racing right why is it
racing because you're worried that
you've got worried that you might have
cancer that's a different thing but the
point is that the feeling is different
the feeling of worry uh distress
anticipatory you know anxiety
well that is characterized by uh racing
heartbeat and and feelings of tension
and so on
but that's the same bodily signature
that you can get when you're powering up
hill on your bike or you know or
something like that so
the level of heartbeat in itself
is as it were you know effectively or
emotionally neutral
what that level of heartbeat depends on
and how it
and how you should react to it well
that's a product of the kind of
emotional context and to understand the
emotional context
that's when you need to bring in the
rest of your perceptual memory and so on
faculties and indeed your

self-representation of you know how i am
functioning in the world and so
to do that
you need to
uh interpret or i just put it here
transcribe those bodily feelings
into emotional feelings of sadness or
anxiety or whatever they are i've given
the example here of um
fatigue because it comes up later but
it's also a good one right
if you're fatigued you know if you're
just tired well that is a certain bodily
feeling no doubt about
but if you're depressed you know someone
who's got persistent depression they
also feel fatigued they feel tired
lethargic apathetic
unmotivated unenergetic
but fatigue in an episode or sorry d
energized bodily dna judge bodily
de-energization in a episode of um
sadness is experienced differently to
just fatigue that's the point right and
why is that because in an episode of
sadness
the world and your prospects in it have
been appraised or evaluated or cognized
choose your term
that is because your your mind has
basically determined that the world for
you is a site of um
irretrievable disconsolation you know
there's losses in there that you can't
recover or recapture
and of course
so um
that produces the feelings of sadness

okay so
why does this so
when we translate this into um
self-modeling you've got the kind of
physical level of bodily functioning and
the kind of homeostatic and signaling
stuff
but then in order to interpret the
significance of all of those changes you
need a model the mind needs a model of
your organism and how it's functioning
in the world and what it should expect
from the world and the kind of affective
consequences that it should get from the
world and so it makes
multi-level um
self-model at the most basic level
at the most basic level you know the
mind models itself
as um
as a unified entity in which all of
these bodily signals cohere
and then it can and use that model to
regulate the body now you can go to
sleep if you're tired at a higher level
slightly the mind models itself as the
entity that has these feelings these
feelings of sin these affective feelings
these feelings of sadness anxiety or
these feelings of happiness pleasure
resentment you know all these subtle
feelings that are modulated by your
actions as you go through the world so
just as you need a model of the uh the
bodily functioning to model the body at
a basic level
to control once to to get through the
world well we get through the world by

acting to manipulate our feelings so we
need a higher level model of ourself as
the subject of affective states
okay and here's the and at the height
it's still higher level still
we report those feelings in a kind of
verbal narrative way you know i'm late
for work you know i'm studying i'm
studying you know i'm practicing
gymnastics you know i'm married etc all
these things well this is just the kind
of the high level narrative self model
that we use to report our feeling we
used to report our states and
communicate them and engage with other
people and so we engage at this
narrative level there's a narrative
level self model that's kind of
linguistic but the narrative level self
model is supported by a effective or
emotional level because as we engage
with other people and report our states
what we're doing is doing that to change
our change the way we feel to feel
better
and at a still lower level those
affective feelings what are they they
are transcribed bodily feelings that is
to say they are emotionally interpreted
bodily feelings
so the um the self model is a hierarchy
is hierarchical a basic level it's it's
probably at a status higher level it's
bodily as interpreted for its emotional
significance and at the highest level
it's narrative
now here's your um here's the hypothesis
about um all these disorders

in a cotard delusion the self model just
disintegrates from the bottom up right
it just falls to bits it falls to bits
at the
physical level right
at in other words this the system
becomes
unregulated and unpredictable
right the the mind is sending his mind
is sending signals control sleep control
appetite control metabolism um control
movement you know get up to it but
they're not getting any traction they're
not getting any feedback
they are the whole kind of modeling
structure is disintegrated
so that
what the mind learns so that the mind
actually can't make a model which
predicts the functioning of the organism
it still has this kind of sensations of
organismic function but they're not
integrated unified and modeled as
belonging to a continuing thing
because and that's precisely because
that model is made for the purposes of
regulation
and when the mind and when the body
becomes unregulated
the mind learns
that there is no there is no model
that's going to predict and regulate its
behavior
so in other words if you think of the
self as a model that the
mind makes at the most basic level for
regulatory purposes
then when it's when it's completely

unregulable
the mind learns that there is no model
that can do the job
okay
we won't say oh
yeah so that's the that's my explanation
of cortad okay
well but in de-personalization it's not
quite like that right because in
de-personalization basically the model
is intact
okay the model is intact you've got kind
of
controllable bodily functioning you've
got cognition perception and so on all
working
but the only thing is
that the affective processes have been
inhibited so this kind of you're seeing
the world feeling the world touching and
engaging with the world but
there's no effective responses to the
world because they've been shut down
for as a dissociative response
so now you've got an intact self-model
but you can't annex or incorporate
effective states into it so you indeed
that's why depersonalization people feel
as though
they feel as though their bodies the
same their perceptual and sensory motor
states are the same
and their narrative ability to report
that think about that you know their
higher level metacognitive states and
narrative reportage
they are intact
but affect is missing

effect has been if you like detached or
taken out of this model that's otherwise
intact and that feels as if
they don't think they feel detached and
on the outside of experience that's why
they say they report they feel as if
they don't exist but they don't say they
don't exist because the model still the
model is basically intact just with this
uh subtraction of its effective
component ability now pain and that is
as a result of unpredicted deactivation
of the anterior insula because anterior
insulin is precisely the structure that
integrates
all the processing
that does those kind of evaluative
emotional processes and produces
effective responses
now in pain a symbolia
which is reported as a kind of
restricted depersonalization that is
actually a problem at the lower level
because pain a symbolic is a problem at
the lower level of the posterior insular
cortex which is more at the level of the
sensory integration
not the emotional interpretation
so in pain a symbolism no susceptible
signals are not incorporated in the body
itself or incorporated in the body
itself model so that's a kind of early
process so
once again in pain a symbolic you've got
an intact self
uh bodily self-model affective and
narrative but somehow no susceptible
signals aren't being incorporated into

that model so and since they're not
incorporated into the model at all
then they won't be transcribed at the
effective levels of self-modeling and
then you'll and then the um
the mind is left with well i'm
experiencing no susceptible signals
but i'm not they're not incorporated in
the kind of bodily self model at the
basic level so
they're not mine so to speak i'm
anthropomorphizing on behalf of your
brain here
and then since they're not incorporated
in the bodily self model they're not
going to be
effectively interpret interpreted at
higher levels of emotional self-modeling
and once again the subject is left at
this kind of narrative level just
reporting on the subsequent experience
it feels like the pain is not happening
to me
okay
and so is this self-modeling theory well
this self-modeling theory um comes from
of course
effective i'm sorry active inference and
predictive processing theories right
uh and this is um from seth and fristen
you know they say that on this theory of
interceptive
this might slide here am i going the
wrong way
on this theory of that that feelings are
generated at the bottom level by this
kind of successive interpretation
in terms of models which are ultimately

top down and within this framework
it's as i said the anterior insular
cortex is specialized
to allow us to integrate information
about body state to feel its
significance
and the point about the self-modeling
theory is
those feelings of significance or
effective feelings are attributed to
a unified persisting entity the self
okay which is the self model i'm just
rehearsing repeating here what i've said
okay
and as they say
there's another way to put it what i've
just said emotion and embodied itself
would have grounded in active inference
of those signals most likely to be me
in other words as you go through the
world
your mind needs to
model the consequences of your actions
and the consequences of those actions
are going to be you know interceptive
internal bodily
and also they're going to be emotional
and affect
and so at any given time we have a self
model which is um predicting
how our actions with the world will make
us feel
and what the kind of sensory
consequences those actions will be at
all levels from
homeostasis body integrity body
integrity and so forth and as they say
the metacog cognitive and narrative eye

or as i've put it to simplify
you've got kind of basic bodily self
you've got and that gets
transcribed and interpreted as a um
effective or emotional self and then um
at the top level we report and
communicate that with the narrative self
basically the self is there to serve as
to to represent the source of endogenous
signals arising from inside and to be
the target of internal regulation
and this uh this goes back and it craigs
papers in the early
thousands you know to put it this way
that you know within the posterior
insulin that does the basic bodily stuff
you know heartbeat oxygenation
energy depletion
and then those representations of basic
body state get progressively and
progressively remapped at higher levels
exploiting uh more more higher and a
modal cognitive processes until we get
to the anterior insula
the anterior insulin has to communicate
with these other systems of perception
memory inference imagination
to tell you the experience to tell you
what to tell you sorry what the uh
the meaning or the significance of all
these bodily changes is and should these
bodily changes are they adaptive in this
situation i'm running my bike uphill
high heartbeat fine
my heart's racing when i'm when i'm
having a conversation with my wife this
is bad
this is kind of anxiety um and so forth

okay i can skip that this is just you
know to say that the anterior and
posterior insula
you know
they are hubs they are not a place where
this they are not the kind of there's
not like a self module a bodily self
module in a effective soft module it's
more that they are kind of integrative
hubs
that basically direct traffic is a good
way to put on or another way to think of
it is they're like the read write head
on a um on a turing machine right
they're coordinating and doing stuff
they're integrating stuff across them
across the brain to this particular task
all right and and they communicate with
other hubs
prefrontal and amygdala which are very
important for collecting information for
the evaluation you know the prefrontal
cortex more about collecting information
for at the country it's a conceptual
level
you know thinking about
you know is my have i have i offended
someone is that why they are you know is
that why they're angry or ignoring with
me is that because of that kind of thing
whereas the amygdala is more like
coordinating information about basic
perceptually driven stuff you know
someone raises their voice or puts on an
angry expression and within a couple of
hundred milliseconds i feel a response
to that so
all right so so where are we

to summarize right these are just a
couple of quotations about the the
insula from various things
let's just read one right the
convergence of multimodal sensory
information and ability to read out
subjective states explains why the
insular is intimately involved in
effective processing correct
okay it's functionally important for
depicting saline information and
interfacing between physiology and
higher order cognitive systems and
that's why it is so functionally
ubiquitous
and connected with so many things to
form a salience detection network it
allows us to feel how things matter
okay but don't within the predictive
processing in active inference it's not
just that you feel salience salience
means
salient to me
relevant to the organism and its
prospects
i just something can't be salient
unless it's relevant to a goal
really relevant to a goal of survival or
salient to a goal of you know
maintaining social relationships or
achieving
um some ambition
so salience is not something in itself
salience is relative to particular goals
of organisming survival or organisming
prospering at various levels
and in order to do that the mind needs
to model itself as the bearer of the

goals as the thing is
prospering or not is affected by its
actions
now here's the thought that
is ethic if you have just like you have
bodily states like interception
you have affective states like anxiety
or depression
and what they what those states do is
they tell you how you're prospering they
tell you how you're prospering bodily
and they tell you how you're prospering
emotionally
which is great and the good thing about
that is
then you can act right you can perform
your active inference of foraging in the
world for solutions of moving around and
sampling sensations and trying out
activities and so forth all these kind
of ways of engaging with the world that
is going to optimize you know your
ability to build a model of yourself as
a functioning system
but you don't actually have to um
fine-tune
all the kind of the mechanisms and
low-level systems
and that's good because we can't right
so if you think about fatigue i cannot
go in and change the metabolism of all
my cells i can't go in and you know
change adenosine levels and i can't
change the balance of serotonin and
acetylcholine across my brain
but i can go to sleep
so i manage the feeling of exhaustion
and then i wake up feeling better having

optimized my um optimize my organism
i optimized my organism by managed my
feelings
and similarly at a sort of higher level
of the emotional functioning i optimize
my organism
by managing effective or emotional
feelings so you know if i feel sad i
withdraw from life and and
avoid the context that make me sad
if i feel happy then i'll keep doing
what i'm doing you know if i feel amused
or interested or engaged then that'll
lead me to kind of pursue and keep
performing the actions that are
rewarding and so on and so on
and so life then becomes a matter of
acting to regulate affect
by active inference as a proxy
for minimizing predictive error or in
other words optimizing organism function
across the
function of the organism so
in other words and the self here is the
thing if the self is modeled as the
thing which sustains these feelings and
the thing whose feelings can be changed
by active inference or action
then that's the role of the self the
role of the self is to be the entity
okay that we can manipulate
okay so in this sense
um
active inference
is like a proxy for overall organismic
regulation
okay so
the self the feeling of being the

feeling of being a unified persisting
self
allows us to get through the world by
manipulating our feelings so the self
model
it's like the airplane icon on a control
panel
on a plane right so the pilot flies that
boeing
737 or 747 or whatever
by manipulating the controls and you can
see on his panel you know the little
model of the plane
banking or going up and down
well that's great because the pilot has
no access to the electronics the
software the hydraulics and all the
mechanisms right pilot can't impose not
going to go back down to the tail and
try and you know change the ailerons or
anything like that and the pilot's not
going to go in there and you know mess
around with the software that of the
computer that controls the system that's
all delegated right that's the way down
all the pilot has to do
is manipulate the icon and that's all we
have to do
we have to just act to make ourselves
feel better
and the self is just the target
of that regulatory actions i'll skip
this now help because i think i must be
finished i'm really running out of time
so
to sum up for our case of the uh the
kotar delusion and these other things
well in the cotard delusion what happens

is you have a version of what these
theorists step on and they'll call
catastrophic dysregulation
so in the case of ndmr
and encephalitis right
cellular metabolic function the whole
thing is kind of
out of whack because of this pervasive
inflammation across the brain and so
there can't be any kind of low-level
signaling that's working and at the
higher more representational levels
you're not getting any kind of
consistent
on accurate representations when it's
kind of converted into computational
language
so
the the brain is disregulable the body
is disregulable so the system learns the
system learns
that
as a regulatory system as a target of
regulation it doesn't exist
of course it's
that is that's just to say the self
model disappears
but you can still feel your body it's
just that those bodily feelings aren't
integrated as
into a model of something that you can
change
and regulate and predict and anticipate
okay so that's called profit
dysregulation so
about that my final slide has got a typo
on it so i'll change that after this but
just to remind you and i'll close off

now because we've been going for too
long but descartes famously said that
it's like hunger and thirst they teach
us that we're not present in our bodies
as a sailor in his ship
but we are closely joined and
intermingled with it so i compose a
single thing with it with it you know
with my body i am my body i feel like i
am my body
we might say within predictive process
that the experience of being the subject
of interreceptive and effective
experiences
produces the feeling
the inference that those experiences
belong to a simple unified thing
and the more that i succeed in
controlling myself by regulating my
experiences my interceptive and
effective experiences the more i learn
that i am that thing you know when i eat
i'm satiated when i achieve a goal i'm
happy and i'm worried and i i resolve my
worries
but states of intractable dyshomeostasis
teach us the reverse
in severe cases like in the ammonium
encephalitis the mind
has to end up learns that there's no
viable regulatory target
because experience is chaotic and
unpredictable so if you can't keep
entropy at bay
back now
and keep entropy at bay at the most
basic level
by having a kind of model that works

then your biological functioning is
messed up and you've got no
physiological control all the signaling
goes to pieces you can't minimize
variational free energy and that is
at the representational level the cell
model disintegrates there's no kind of
self model that predictably
enables you to regulate yourself and
you're just left with this kind of
chaotic experience in the cotard
delusion which you report with the kind
of the shadow
that's all you've got left is you know
linguistic functioning
um
[Music]
you've just got linguistic functioning
rooks
okay you've just got the reporting
talking
so that one cotard patient famously said
ah
i'm just a voice and if that goes i
won't be anything
if my voice goes i'll be lost and i
won't know where i've gone so
the patients this is the patient's only
response to this intractable sense of
bodily effective entropy the linguistic
shadow the narrative eye
completely disconnected from the
structure of bodily ineffective
experiences that normally anchor it
because the small model has fallen to
pieces and
i should stop there because i've been
going for a long time

thank you you can
share the screen and we can
talk a little bit it's just as long as
you'd like
yeah sure same button as to share and
then you can
cool
i wrote down a ton of stuff there were
some questions in the chat so thanks for
the awesome talk
i'll start with one of dave's questions
so dave had written
does your evolutionary and neurological
exploration of delusion especially of
the pathologies of the prefrontal cortex
functional connectivity
preclude the validity of psychological
approaches
or how do psychological and psychosocial
approaches benefit from a corp
incorporating these kinds of methods and
findings
well that's a very good question i i
don't think they preclude at all right i
think they're i think they're vital to
your psychological approach i mean i
didn't i did write a it's a chapter in a
book i wrote a while
about this but
i don't there's a bad tendency in
psychiatry in the world to think okay
you've got psychological approaches
you've got people's thoughts their
experiences and feelings and then you
have
the kind of things that biological
psychiatry does you know molecular level
interventions with receptors and

serotonin and that they never they never
talk to each other right and so you have
a disease model of mental illness
and you have a psychosocial model of
mental illness you know this
but clearly that's wrong right
and it's got to be wrong because if you
think of um think of a just an example
like astigmatism
or uh you know macular degeneration
that's something that you experience at
a personal and experiential level as you
know loss off part of your visual field
if you've got a black spot in your
visual field that affects how you
experience the world how you get around
it how you interact with other people
and there's a mechanical explanation
there which is in terms of
you know um
the visual visual cortex or the retina
or whatever and the capillaries and all
that in the
okay
well why why would it be the case
and if someone said to you oh well look
you know we could only have um
like a personal level an experiential
phenomenological account of
uh of visual experience
you'd say that was mad
it's oh i want to go to the eye
specialist and he knows all about you
know how diabetes produces this thing
because we have a mechanical thing but
not only that
the mechanical explanation
depends on a computational explanation

of how vision works
these cells in the retina get bombarded
by photons
they
regret and the mind progressively
reconstructs
um an image of the image of the causes
of that bombardment and that's our
visual experience
so i can tell you why macular
degeneration you know at any level at
the level of the retina or visual cortex
it's going to affect your experience
and we think that's all in order and we
think that's exactly the way to do it
why wouldn't we think of that for the
kotar delusion or for um
or or for uh you know any psychiatric
disorder anxiety or whatever it's not
that
you've got this kind of
autonomous level of um
autonomous level of uh you know
molecular level interventions and then
you've got some phenomenological level
of experience and thinking
i mean one depends on the one depends on
the other
i just think that's like that
yeah so so in my view
the right what we're trying to do is
is try and understand what's happening
in the intervening level and that's what
theories of visual processing do you
know are you got artificial theories and
you've got neurological theories and
they come together and that's what the
active inference self-modeling theory is

supposed to do in these cases of
pathologies of self-awareness you have a
kind of experience of yourself has
changed
and we know that that's because
something is happening in your body and
brain
how are we supposed to explain that and
not just correlate the change in
experience with the
low level cause
we want to say that's because these
changes in your body and brain are
organized to do this information
processing kind of computational
representational functions
and when you get these low-level changes
they compromise this functioning
this computational level working
with the result that you have these
experiences and it goes the other way
you know when someone you get
information from the world
it has to filter all the way down
by being represented processed somehow
and you know we have to take a
stand on the nature of the comp of the
nature of the computation whether it's
wide narrow symbolic
neural network whatever you know that's
ongoing project so that's my it's my
thoughts on that topic
thank you so the next question is from
dean who wrote
does this does the disassociation of
feelings
leave the person not able to make a plan
because in some way their predictive

model has been compromised so how do
these people you know do they grasp
objects appropriately could they get
across town on the bus what level
or in what ways is their action
prediction
changed
but yeah well i think you know i i think
people who are very effectively
flattened right and depersonalized
uh i think planning is an issue planning
is an issue for them
um precisely because you know
of the idea that you know if you're
trying to decide where to go for
holidays or whether to get married or
whether to quit your job
it's not just a sort of cold cognitive
process of weighing the costs and
benefits
those things have to be felt
right you know if you decide or am i
going to go to hawaii where it's warm
and pleasant or am i going to go to
alaska and just get well how those
you mentally rehearse those options
and how that makes you feel determines
the decision you know and if you decide
to get divorced or get married um
it's not just oh yeah you know if i get
married this and this and this will
happen and if i get divorced the
likelihood of this it's more like that
makes you feel a certain way
so if your life becomes effectively
flattened so that
perception and cognition of all the
various

options for action
are now effectively neutral
it does make it very hard to uh to plan
and engage and and in the end i think
what you find in depersonalization is
people just kind of go through the
motions with their habitual actions
you know so if you're a bus driver or a
university lecturer or a husband and
parent
then but you feel depersonalized then
you just you keep doing what you're
doing
but you feel like uh
just the empty shell going through the
motions but those people had a lot of
difficulty
coming up with the coherence
alternative plan it's like why don't you
do something else why don't you you know
but but we see this actually you know
even in um
in everyday cases of life you know you
might know that it would be good if you
took up exercise
but it would be you know as a objective
piece of knowledge
it would be great if i took up you know
bike riding or if i went on a diet or if
i you know whatever if i moved to cities
but
but you just know that as a kind of you
know piece of information
it won't make you do it
unless that somehow can click into your
affective self-model it's like yeah it's
me
i can kind of imagine myself as you know

whatever it is riding up a hill walking
up a you know having some future that i
can actually inhabit
not just intellectually but also
emotionally and personally
well just one thought on that and then
i'll read a question from dave is like
in a way society
is a big top down like in active
inference we talk a lot about nested
systems and you focused on sort of the
level of bodily integrity and brain body
but in a sense the social
structure our niche scaffolds our
behavior whether it's hey it doesn't
matter if you don't like broccoli you're
gonna put that fork in your mouth
or whether it's some other type of
decision
there are top down priors that actually
structure
our decision making and planning
in the absence of precision or of
a good way to plan
so just really interesting there
i think that's exactly right and you
know um
i think this is really your point
exactly applies really really well to
addiction
there's a woman called hannah pickard
who writes extensively on addiction and
what she points out is that
an addict can know that if they stop
taking drugs it would be better for them
but so what you know what they need to
be able to do is project themselves into
the non-addicted future where they're

living engaging in the world and they're
getting positive feedback from the world
but she
says to pick up your point
that's very hard to have a model of
yourself like that
without social scaffolding without
social structure without a kind of um
positive or without a kind of positive
feedback
and structure from a world around you so
the thought is the the self that you are
is is as you say your priors your
predictions
they're not just about internal modeling
they're also about the price
expectations
and plans that and plans and
reinforcement that you get from being
immersed
in a sort of social media and for
addicts that's not there
they can't imagine they have no model of
themselves as functioning in an
alternative social media so when they
get out of jail or get out of rehab
unfortunately they get back in their old
social media
they're familiar triggering situations
in the familiar social environment
and that kind of activates their
self-model as someone who
who does this does that
and there's and and she says when you
ask them to reflect on
what it's like as a non-addict imagine
they can't
so there's a kind she said it induces in

them a kind of existential terror
because um they really
can't
imagine themselves
as a non
non-addicted life it's kind of it's an
empty future for them
partly because she says there's no
viable social scaffolding to support
their
their life in that world so your point
is i think
plays really well into that that
literature
and what you said there um what is it
like to be you know in that case not
addicted to a substance or a behavior
it's kind of a subtle allusion to the
famous philosophical tradition of what
is it like to be a bat to be an ant
colony et cetera
and when you brought up i think even at
the beginning you said what is it like
to feel you don't exist
that
this idea of negative space
and of negation came up in such
interesting ways in the talk like if you
ask somebody what does it feel like to
exist it's like that's 99 of the
discourse or you know that's the mystery
but then by drawing our attention well
what would it feel like counter factual
importance for active inference what
would it feel like to not exist as like
i know i'm not feeling that so what is
that one and then what's the other side
of the coin of feeling like i don't

exist it's like it's like feeling like i
am in control like i am steering the
ship so that was so such a cool
connection
that's that yeah i think that's the
point i wanted i wanted to to get at you
know what why i like the active
inference framework for this is because
the feeling of being me
you know a lot of people tell you oh
well if you have a perceptual experience
then someone's having it you know
an
experience has to have an experience or
something like that
or how can you not feel like you're me
if you have a body you can see your body
touch your body and feel it but that's
not what we're talking about the feeling
of being me
is the feeling of being in charge
the feeling of being the thing
that i can manipulate
in order to bring about all these
experiences
right and you can still have experiences
right because plenty of experiences
but if you lose the feeling that those
experiences are things that you can
change and manipulate dependent on your
intentions
then you lose the experience of being
you and that's at a high level but the
same thing imagine the thing going right
down you know really low your your
hypothalamus is sending signals to raise
your heartbeat and not getting any
feedback

your heartbeat stays low stays
obstinately low
or obstinately high it's the same thing
i'm not i'm not getting any traction
here
obviously a hypothalamus doesn't think
that but
you know
or you know imagine a bacterium that's
trying to contract its membranes to to
turn
left to pursue a sugar gradient and
nothing's
happening it's getting keeping getting
this kind of obstinate feeling that i'm
not in charge anymore
and i've got no model that i've got no
model on myself that's working to
to to give me the feeling of being me
and that's
in such a case the system still has
sensory experience sort of activity at
its receptors and stuff is happening but
it's just not
a regulatory target anymore
and and that makes me think about um
when you talked about the people who are
suffering from pain but they're
disassociated from it it feels like it's
not their pain
it's a little word play so it's not a
philosophical argument but gnosis
and nociception
it's like there's the knowledge of pain
but actually it's almost separable
from the suffering
and so this is an extreme case where the
gnosis of nociception has been entirely

separated from the narrative of the self
it's like yeah there is pain in the arm
but that's not me or it's not part of my
narrative and so it doesn't bother the
person psychologically
that's exactly right
there's damage there's damage to the
body that i inhabit
is that's what the brain is saying
by this body that i live in is damaged
but it's not my damage
now do they ever ask who's is it or does
it just not seem to be like oh it's just
like oh someone's car someone's parked
outside or is it like i'm kidding
no no um
the really interesting thing i mean the
early cases the really interesting thing
about um symbolics is that you know
they don't take any uh action to avoid
pain
you know so that in in that sense you
know that's really bad and so but it's
it's not quite the same as um congenital
insensitivity to pain which is a problem
with the receptor level right those
people don't detect pain
and so um
you know they're in a bad way they put
their hand on a burning stove they um
break their legs and they don't they
just don't feel they don't even have the
bodily sensation
but pain asymbolics they have the bodily
sensation
but they don't
annex it to themselves
well it points to different um roles for

different receptors so i want to ask
about the nmda receptor which you kind
of brought up in the beginning what does
the nmda receptor do
neurologically or why is it associated
with that phenotype
right well now you're um now you're now
now you know now i'm up now i'm out of
my league um
but uh it i think that that kind of
encephalitis is a um inflammatory thing
right it's an inflammatory uh response
and so
when it hits those receptors i i think
it's just because uh ndma is like it's
so widespread throughout the brain
i think it's like uh
i'm talking out of turn here right so
it's not it's not my my field so cancel
this but
i think it's like like a like a catalyst
or modulator for so many metabolic
functions
right it's involved in everything as a
kind of very fundamental thing so when
you mess with it
you pretty much
you that's why you get this kind of
catastrophic
dysregulation across the brain
ah yeah that's that's the thought it's
one of these it's like um
adenosine or something like that you
know that these uh metabolism depends on
that and if you mess that up or uh
what's another one uh
the ketones
you know if you if you get a buildup of

uh phenylketone in the brain
you know it's this tiny chemical but if
you tiny levels of this chemical can
just destroy everything produce mental
retardation mutism
motor disorders it's a catastrophe
an absolute catastrophe because
you know
the cellular level functioning is so
dependent on these nano nano level
adjustments of these basic things
that
speaks to two dimensions of the
challenge which is the multi-scale
systems there's little things like
molecules and big things like people and
then also the trans disciplinarity and
it's like we should have one story for
one level from one discipline and then a
separate discipline and a separate story
we need something like an integrated
approach what's that going to be
that's exactly right and so where i
suppose where i came into this when i
was you know when i got into philosophy
and people thought well that what's the
integrative story going to be well you
know some kind of computational story
you know
top-down classical computational
modeling cpu
modular inputs modular outputs that's
one way it's a kind of broad brush
approach then you have neural network
stories
and then you have you know deep learning
stories and all these different
architectures and now you have and they

have active inference
and it depends what you think uh active
inference is gonna um how it's gonna
play out i mean if if you think of
active inference as
you know at the biophysical level it
sounds right but it's very abstract
right it treats like a human organism as
like a physical system in space
it's like a self-organizing asteroid or
something like that is really
the same the same laws of physics apply
and then you have the kind of bayesian
implementations of
active inference
it's an interesting question you know
whether you think the brain is actually
um
bayesian
you know because a lot of psychologists
will tell you that
the brain departs from bayesian optimal
reasoning almost everywhere in almost
all and almost always you know
so yeah
very interesting and welcome back dave
so um one of the questions from dave um
was have you looked at functional
delusions
socialized as in mobs or depth
psychological
no no i'm just i'm still on the um
i'm still on the
standard curriculum of um
delusions of misidentification cotard
delusions schizophrenic schizophrenic
delusions i haven't talked about maths
psychology and depth psychology i had

some thoughts but
i mean clearly like self-modeling
applies right uh if you're talking about
how ideas proper propagate
in populations
then i guess you could say
those those ideas which are most
consistent with a self model
they're going to be more easily absorbed
and
stable stabilized you know within the
individual
so it's going to be very hard to get
traction for an idea or an ideology that
doesn't fit with a self model and
probably
self models are
since they're constructed to
regulate affective states
that tells you that the ideas which
propagate and stabilize are going to be
those which
get traction with people's um
effective
self-model you know what that make them
feel a certain way
rather than a kind of abstract
intellectual argument you're not going
to persuade an anti-vaxxer
by giving them statistics
and you're not going to get a kind of
propagation of a wave of um something
across social media on the basis of
you know an irrational inference
but but that's i don't have anything
super much to say about that
very true that we see sharing and
activation patterns

in the sort of global brain based upon
affect not veracity or accuracy or
any other feature
um and and maybe that speaks to another
one of dave's questions which was
how does your work illuminate the notion
of cognitive dissonance or consonants
even in his technical work leon
festenger seems not to succeed in
building a credible dynamics of
compatibility and incompatibility among
psychic atoms
relevant passages of gregory bateson's
essays in steps towards an ecology of
mind seems similarly limited so whether
or not you've read those specific texts
like i haven't read those specific
things i suppose one way of thinking of
it is
within the kind of active inference or
within the um
predictive processing framework is you
know if you get some discrepant
information
that uh
contradicts yours contradicts a
self-model or contradicts a predictive
model you have you know some choices
you can revise the model
that's you know perceptual inference as
they say
or you can explore the world and try and
get some you get some further
information to confirm your model you
that's that seems to be the active
inference approach and obviously life is
a balance right life is a balance
between the two

keep your model or get some more
information
but it seems like if you if we take the
scientific uh
analogy of you know hypothetical
inference
often what people do with discrepant
inference in sorry discrepant
information is just compartmentalize it
you know they just put it away
and scientists of science you know known
through it no one throws out a good
scientific theory just because of some
discrepant
evidence
even if that can't be reconciled you
know right now no one's chucking out the
theory of evolution because someone came
up with a wacky organism or a wacky
function you know it's like hmm there
must have got to be some explanation i
don't know what it is but let's just
we'll just keep doing our thing
and hope that eventually it'll be
realized you know so i don't think
people
these ideas of kind of um
updating or upgrading hypotheses in the
face of flux of new information that's
actually not true because we can't be
that sensitive
to information that would be mad
right
so
and i guess if you have a certain self
model that's getting around the world
and functioning
you know if you're a religion if you're

a highly religious person or something
like that and you live in a religious
community and you go to college and you
know
read dawkins or something like that
well i mean you have a choice you can
throw out your whole self model in your
whole worldview
or you might just think like many
surgeons scientists
and otherwise rational people okay this
is my life
in the hospital where i'm a
neuroscientist or whatever
and this is my life at home where i'm a
preacher and a
patient family patriarch of a religious
community
um
because complete consistency across all
the levels and all the functions
is
too much but i would say because it
threatens a kind of self-model it's not
because it's um
you know that kind of
compartmentalization is not so much a
matter of intellectual consistency so
much as
consistency with the self model
and very importantly that the self model
includes the sensory evidence and that
meta model so even though my background
is in evolutionary biology
very early i realized that
people who believe in evolution people
who believe in creation every other
flavor look at a tree and see evidence

for their world model so that's the
starting point for the discussion not
that you see the evidence but that
everybody sees evidence confirming their
worldview and that's a very different
discussion than the dogmatists who might
be on any side
on the other side right and yeah and no
one is going to come no one's going to
come to the top no one's going to come
to the topic with no model
that's going to say put aside your
preconceptions well there's no such
thing right so
uh yeah yes definitely and that is
relevant for the uh cognitive bias and
ai bias discussion
in these future systems but we'll have
an unbiased system
what is that even going to mean
yeah exactly and that is the you know
the strength weakness of the bayesian
thing isn't it because
the bayesian idea is you know you start
ab initio and then you get some evidence
and you update your model you get some
more evidence and you update your model
and it's on but you've got to start from
somewhere
and so much of the action in bayesian
theory is in the priors
but there's no theory internal to base
which will tell you where your prior
assumptions about how much weight to
give incoming evidence
is going to come from
it's a little bit more like blockchain
with the way that we can make our priors

explicit
you don't step around this issue by
using frequentist statistics or by
casting tea leaves so at least we can
have a discussion hey i use different
priors i got a different answer versus
we use radically different families of
priors and hyper priors and we're
converging on a similar action
so maybe that speaks to at least this
action being appropriate if not it being
the ultimate yeah that's a that's a yeah
that's a very good point it's not as
though um yeah it's not as though anyone
else
has a solution
to uh to where do we start from you know
what what are our foundation what are
our foundational assumptions um yeah so
it's a bit rude to criticize
uh bayesians for saying there's there
has to be a lot of action in the prize
right there has to be a lot of
information in the prize to get the
project off the ground
um
true
but you know they're the same with
anyone you know chops get chomp skins or
anyone
they just
yeah
exactly
so one other um
phrase or piece that came to mind was
almost like parametric hysteresis
the idea that although most sensory
evidence is sort of just sloughed off

kind of like teflon on our mind you look
away it's just totally gone
there are traumatic experiences and then
on the other side there are um
healing modalities whether it's a
healing experience or a biochemical
intervention
there are sort of like
there's interventions
that
restructure
our models and we can think about that
in active inference perspective so i
just wondered like philosophically or
from any view
how do we think about these kinds of
changing events whether kind of changing
in a negative way like a trauma or in a
positive way like healing
how do we scaffold for that
well that is actually a really good
question because you know in this in the
case of psychiatry it seems like um
almost all therapies are equally good
and equally bad
like none of them really none of them
really work in a way that we can kind of
predict or explain there's only a
statistical you know
statistical correlations some people
took cer antidepressants and you know
certain proportion get better but then
they relapse some people went to a
therapist or cognitive cbt
some get better some relapse you know so
what kind of what kind of interventions
work
is really uh interesting and the kind of

restructuring you know and that's the um
that's part of the motivation i think
for uh psychedelic uh approaches to
psychiatry where it's like okay
one thought might be look i can't we
can't really have a successful
intervention because that would require
you to rebuild or remodel
yourself it's not just changing your
perception and changing your experience
it's the way you incorporate that
in your whole kind of regulatory
framework how we're going to do that
well maybe with 10 years of therapy and
you know lots of um patient
reconstruction or or sometimes a life
changing event
people give up drugs because you know
for some reason
it's a life-changing event or they just
grow out of it people say
but it's hard too
hard to systematize that isn't it it was
psychedelics what people are saying is
well let's disrupt their self model
let's kind of for a while let's
introduce maximum entropy into the
system again
you know
so that
they experience life
briefly hopefully safely
uh
with a lot of their self-model um
disintegrated
yep and so
they get to see life for once unfiltered
you know normally everything is filtered

how does this matter to me in relation
to me in relation to my history my plans
you know the other people that matter to
me
all of life is you know just being
filtered at every level like that
and that's very hard to to experience
life without that filter
and
psychedelics give you a um
a brief
immersive very destabilizing
experience of life unfiltered
and maybe from that can be a starting
point for something that's good
classic william blake doors of
perception
um idea and also the the rebus work by
fristen and and carhartt harris as well
as adam saffron and actually asking like
maybe there's certain aspects that these
neurochemical interventions they
strengthen maybe it's not just this
across-the-board weakening of our world
model but there are certain levels of
analysis in which things might be
strengthened and right right
on this note though of the the treatment
and the recovery one of the very early
slides i think it was mr a
um was the patient and then um
came in initially presenting some
phenotypes and then there was plasma
exchange
and it said after a few plasma exchanges
you know he had improved and then he's
had this other um thing and so i thought
um

interesting treatments
what does it say about the body and the
brain or what made those doctors go for
the plasma exchange and was it with with
a matter
right well because um
because
they had this idea that uh
they had they had this they had this
idea that inflammation
you know was the core of a lot of these
catastrophic the core of a lot of the
symptoms when you when you look at
depression and you look at
psychiatric disorders people often get
wrapped up in the um sensory experience
and the um
and the cognitive experience you know
people say things that are so crazy
you know
and the thing is why would you say that
why are they so irrational
you know and people start at that level
and they talk to them and they say
try and understand their life experience
and all this or whatever
but the view of these uh guys in mexico
was look you know look look at the other
symptoms
um
[Music]
look what's happening you know
the guy is pissing his pants
himself hasn't been to sleep
for three days and then he sleeps for
two weeks
and he doesn't want to eat anything he
hasn't eaten for a week you know so this

is like really basic
it's really basic functioning is out of
control now um
and that looks like it maybe that's a
kind of and we know that uh you know
inflammation does affect homeostasis
really badly
so that was their thought you know let's
um
and then let's test them for all these
you know
antibodies for encephalitis and they
find it so plasma exchange is the um the
treatment for that
and then of course you might think oh
well all right but
that's not something you're going to
repair straight away
you know when you've gone from
homeostatic catastrophe
to uh
a bit of reintegration
you know you're still going to be left
with the cognitive level has to get back
in touch with
you know if you're thinking of those
kind of levels of organization and
systematic functioning and interaction
well okay
you change the molecular level function
you don't change everything straight
away which is why um
it's a very nice study by katherine
harmer you know give people
antidepressants
or anything like they give people
antidepressants or you've changed their
serotonin levels within an hour

right so you brought their serotonin
level across the brain back to now why
don't why if sir
the serotonin theory of anti-depression
is right why don't they feel better why
don't they change their pattern of
thinking
it does affect
it does affect some very kind of low
level perceptual processing though so
they're patterns of gaze
you know depressive people their circads
are very avoidant you know if you show
them emotional faces or something they
don't look at the the nose the eyes the
mouth
their eyes automatically avert and
people with anxiety and so on so they're
not getting any positive uh you know
emotional feedback from their social
world
but if you give people antidepressants
that pattern of cicadas changes within a
few hours
so it's almost like you're giving them
the ability now to um
do some active inference and reacquire
some some some new information about
about their world
so if they go around the world luckily
and they people smile at them and laugh
and talk with them then they they're
getting some new information but
that's not going to disrupt
straight away all the high level models
that they've built
to um you know you're not going to
change their that's not going to happen

straight away
and that
um story about the serotonin story i've
personally felt
is that disciplinary hierarchy it's like
well we know how this drug works look at
my simulation of how it binds to the
receptor you bring up but then if it did
that and that's simply how it worked if
it weren't just um you know changing a
complex biological system
then wouldn't the time course of action
be very different
and exactly it's a it's a um
looking where the light is because we
can make these simulations with a
docking of a molecule
but as yet
we lack some of these layers in between
absolutely the vibrating molecule and
our own phenomenology which is sort of
that fog of uh
experience that we're exploring
together
yeah i know i i absolutely 100 agree and
i think um
you know yeah but
most of my practicing psychiatrists
colleagues and you know collaborators
they would agree too
but there's a big uh pharmaceutical
industrial
complex
you know that has that has a interest in
thinking you know persuading people
otherwise and also people are desperate
yeah yeah
can't wait till we have big prior

and big precision
yeah yeah exactly
you you brought up the um
australian philosopher colin klein
who
i was familiar with because of the
collaboration with andy barron
actually about basil experience in
honeybees and in insects
so just was an interesting connection
with thinking about okay if we're going
to take this sort of bottom-up
computational approach then
where we take that in not just the
clinical but in the evolutionary context
yeah no he's a great he's a great uh
philosopher for sure and he's really got
his finger on the pulse of all the um
i think
cross-disciplinary
um
yeah material that you need to do that
kind of thing well he's really good like
that
here's another just in these last few
minutes a question from the chat
cb wrote
any thoughts about the role of
non-synaptic information transmission
across the brain
especially
emphatic coupling thank you kindly
um
honestly
[Applause]
i don't really know um
yeah i don't because uh
clearly it's got to have

there's no reason within say active
inference framework or actually any
computational framework
to think that um
you know the the fundamental mechanism
has to be synaptic
uh
although
it could it could be anything right
how the brain
is organizing
uh the body to get through the world to
survive could be anything it could help
itself to any mechanism you know
astrocytes glial cells whatever the hell
you know
but we have got
we bought into this idea that it's uh
synaptic
probably going back to heb and people
like that and a neural network theory is
that it's a synaptic theory right you've
got these units
uh interconnected units with weights
which are supposed to model some
abstract levels synaptic functioning
right
we might be off we might be off um off
track there
but
so maybe this is a kind of a good
advantage good advantage of active
inference it's like
at the moment uh computational theory is
hung up on the synaptic model because
that's what deep learning and neural
networks are
right setting of setting weights across

the brain but that's a very abstract
idea of um
heavy and you know it's a very abstract
idea of what ed was on he was on about
and most neuroscientists like i had a
guy talking in my class the other day
invited him to give a lecture he works
on dragonfly vision
and he was just talking about the
neurons in in the eyes of dragonflies he
was saying there is all this stuff in
there that we don't know
glial cells astrocytes what the hell
that you know what they're doing what
they're encoding how they're encoding it
but it says if you mess with their
function
then you mess with the ability of the
dragonflies to to
do their incredibly accurate flying and
landing and sensory representation so
yeah so maybe the synaptic model is um
once again it's a case of looking where
the light is you know
artificial neural networks
they're great they can do a lot of
things
um but
they are a kind of high level
abstraction
then again bayesian models are an even
higher level abstraction
if you if you want to put it that way
because the bayesian mod the bayesian
models just say okay you've got a system
interacting with its environment and
i've got a function
a function that tells you

how the system's going to interact with
its environment
and i can devolve that into a lot of if
you like i can devolve that into a lot
of subsidiary
uh
levels of
of bayesian
bayesian inference all the way down but
how do they how do they glom how do they
glom onto
the actual molecular soup
uh
i don't know but i mean that's the
that's the job of it isn't you know
that's you know that's why i i guess
still
uh dopaminergic uh functioning in the
brain is still a bit of a triumph
because
sorry understanding the dopamine system
is because you have
a system whose properties
map pretty well to the predictions of
formal theory
you know of temporal difference learning
and reward learning not perfectly but
you know
it does fit
so that's good so that's good you know
that's good it seemed to seem to say
we're in the ballpark
a paper on the dopamine signaling about
explanatory pluralism from a
philosophical perspective about
the way in which the dopaminergic
signaling um in in different ways can be
seen as consistent with the precision

account the reward account or other
accounts and so it's like
maybe let's um you know wait a few
decades yeah
before we um you know give out the
trophies let's yeah yeah yeah absolutely
absolutely so
i'll just very good yeah because that's
right why why can't it doesn't fristen
have a paper on that
even you know and actually there are a
couple of good papers on that now
yeah with the within the active
inference framework we can conceptualize
the um the dopamine system as you know
fine-tuning fine-tuning patterns of
active inference
rather than back propagating a reward
signal
exactly
exactly yeah so i'll just close with
this sort of um this diptych of
questions a little high road in a low
road one of the paths would be what
research directions do you think are
exciting for people to pursue and the
other side is um how do you think for
potentially non-researchers
what you've spoken to today might
change the way that they think or act
[Music]
i think for
researchers i think i think any
researchers should still keep
researching you know what they're
interested in
like if you've got a fascination or an
interest there's dragonfly vision

dopaminergic signaling whatever i think
you should just keep going for it
and incorporate uh
active inference or any framework that
that is is
is helping you know i don't think
jumping on a bandwagon or any researcher
is
helpful
although
you've got a bit you've got to bear in
mind that if you're trying to function
in an institutional environment you're
going to be on some bandwagon right
you're going to be in some lab or some
in a school of thought but
i think you still should you've still
got to do what you're interested in
rather than promote
you know
someone else's program so i don't have a
kind of
i would love to see
i would love to see that um
that you know kind of fristonian ideas
and everything could
be put in a
intuitive way
that made them more accessible
for people who were looking for a kind
of a kind of recipe or something like
that
because they're so technical and they're
so formal
and uh it's hard to extract um
how to extract the gold you know if you
know what i mean it's it and that's why
yeah so

i think
so a good suite of examples would be
really helpful
like you know standard cases that people
use to explain it
um
would be good you know just as you know
often these examples the examples are
wrong right so
you think of mars theory of vision
okay probably wrong
but at least it kind of gave you a way
to understand
you know what's going on in the visual
system and then you can fine-tune it and
change it and argue with it and
everything like that but he gave people
a way of taking the understanding of
perception out of this out of you know
out of a really kind of technical
specialized
framework and now philosophers talk
about it psychologists
it's probably become a little bit of a
um an albatross around the neck of
because but no one's come up with no
one's come up with like a
that one it would be nice if um someone
could
i mean because the ma framework is so um
so so clear right and and it works for
active inference too you can say well
look
the mind is in the business is what holy
says of um
reconstructing the causal structure of
the world you know inferring the causal
structure of the world outside beyond

its boundaries so it's very internalist
view
but that's a very um plausible way of
doing it you know i think yeah i get it
you know you can explain that to
students and to people look
the brain is stuck inside its skull it
can't it can't engage with the world it
can only ever do it by proxy indirectly
and what does it have photons sensory
receptors
it's got to build the picture okay and
you know that's what ma said and and how
he put it in the predictive processing
framework
something like that would be helpful if
you get that idea across you can do it
for other modalities of course
interoception you know whatever
sorry what was the the other question oh
on the non-research frontier is um just
in the day-to-day how does this
influence how we might think or act just
in our day-to-day experiences
i think it makes a big difference if you
think of yourself as um you think of
your behavior
as uh confirming a model
you know i think it makes a really
really big difference you know if you're
arguing like you're arguing with your
partner or
you know stuck in
a way of life or ways of especially
especially social interaction you can
think of social interactions you know
i've got a thought i've got to try and
get a point across or i've got a goal i

want to achieve the goal
right i want to make friends with this
person i want to do this i want to do
that
um
and then you can get pissed off or
whatever if you're if it's working or
not but maybe the right way to think of
it is you know
i've got a model of how i function in
the world
and uh you know my brain has built this
picture of how i function in the world
and i'm just now
i'm bungling around the world trying to
optimize it
you know so what i think what i think of
as this kind of you know
autonomous engagements of this um you
know
individual who's above it all and
choosing and whatever and you know
making decisions and
not really maybe what i'm doing is
bungling around the world trying to
optimize my model
and um maybe i've got i can get stuck in
a local minima
and i can get stuck in a
you know all my models my model's not
working we had this discussion maybe i
don't know if you know this philosopher
dominic murphy
he's a fantastic philosopher of uh
science uh science and psychology
but we would think of you know this
national model and you know
a lot of the time what you're doing is

uh brand management you know yourself
who you are
is this model that you've made that
works for you in your environment your
family as you're growing up or high
school or whatever like this and you've
got this brand now and unfortunately
or fortunately that's how other people
see you
your actions are really about
getting your brand out there brand new
brand management and if you identify
with your brand
great you know
but if the brand if your brand isn't
working for you
and if you and if you just see your
actions as oh god
i am just what i'm doing is kind of
finessing and thought and reinforcing
and fortifying
this model of how i act in the world
does it have to be like that
you know
very
nice point that instead of um social
interactions maximizing my reward your
reward our shared reward zero sum
we can actually think about well what if
we were both reducing our uncertainty
about a positive social interaction
[Music]
what might be known as win-win in the
reward paradigm but here yeah that's a
really good way to it's a really good
way to think of it that's a really good
way yeah exactly
thank you for this awesome visit it was

really uh fascinating talk and one of in
the chat brilliant practical advice
thanks daniel and thanks for your guest
so yeah
thanks a lot from our lab and you and
anyone else are always welcome to join
anytime
thank you very much it was a really
great experience nice to try these ideas
out and this discussion was really
helpful and informed you know so thank
you very much for the chance and i'll be
um propagating
you know i'll be spreading the news
about the lab so thank you thanks a lot
thanks a lot bye see you everyone bye
nice talk pretty cool